Mackenzie N. Pitts

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EDUCATION

PhD, Mechanical Engineering – University of Washington — Expected Graduation: June 2027

Dissertation: Personalizing Clinical Gait Assessments for Children with Cerebral Palsy

MS, Mechanical Engineering – University of Washington — 2024

BS, Mechanical Engineering – Auburn University — 2022

Dean's list, all enrolled semesters

RESEARCH EXPERIENCE

Neuromechanics & Mobility Lab – University of Washington — Sep 2022 – Present

Advisor: Katherine Steele, PhD

Leveraged wearable sensors and machine learning to reconstruct inertial signals of gait [P1]

Built a data processing pipeline in R to extract gait mechanics and output participant summary reports of motion capture and electromyography measures

Applied Bayesian principles and time-series analysis to personalize oxygen consumption [C3] and gait asymmetry [C1, C2] tests for children with cerebral palsy

Created data analysis tools to disseminate in clinical settings for gait assessments

Graduate Engineering Education – University of Washington — Sep 2025 – Present

Advisor: Michelle Hickner, PhD

Evaluated the effects of educational background and interests on learning in differential curricula

James R. Gage Center for Gait and Motion Analysis – Gillette Children's — Spring 2025

Supervisors: Alyssa Spomer, PhD, Andrew Ries, PhD

Developed a generalizable Bayesian model to estimate oxygen kinetics during walking

Collaborated with physical therapists and clinicians during patient visits

Performance Research Lab – University of Washington — Sep 2022 – June 2024

Advisor: Cristine Agresta, PT, MPT, PhD

Investigated the effect of midsole material in running footwear on center of pressure

Advanced Materials – National Institute for Aviation Research — Summer 2021

Supervisors: Keith Roberts, Joel White

Analyzed mechanical properties and failure modes of high temperature polymer matrix composites to develop a new ASTM-aligned test method

PUBLICATIONS

P1. **Pitts, MN**, MR Ebers, CE Agresta, KM Steele. Evaluating Sparse Inertial Measurement Unit Configurations for Inferring Treadmill Running Motion. *Sensors* 2025, 25, 2105. 10.3390/s25072105.

P2. Ebers, MR, **MN Pitts**, JN Kutz, KM Steele. Human motion data expansion from arbitrary sparse sensors with shallow recurrent decoders. *bioRxiv* [Preprint]. 2024 Jun 3. 10.1101/2024.06.01.596487.

CONFERENCE PRESENTATIONS

C1. **Pitts MN**, A Spomer, AJ Ries, MH Schwartz, KM Steele. “Comparing split belt treadmill adaptation rates between children with cerebral palsy and typically developing peers using Bayesian models.” *10th World Congress of Biomechanics*, July 11-15, 2026, Vancouver, Canada. *Podium*.

C2. **Pitts MN**, A Spomer, AJ Ries, MH Schwartz, KM Steele. “Mechanics of Gait Adaptation in Children with Cerebral Palsy.” *Dynamic Walking*, June 22-27, 2026, Seattle, WA. *Podium*.

C3. AJ Ries, **MN Pitts**, KM Steele, JM Donelan, MH Schwartz. “Accurate steady-state VO2 estimation with less metabolic data using calibrated patient-specific Bayesian regression models.” *ESMAC*, September 14-19, 2025, Maastricht, Netherlands and Hasselt, Belgium. *Best Paper Nomination.*

C4. **Pitts MN**, A Spomer, AJ Ries, MH Schwartz, KM Steele. “Mechanisms of Split Belt Adaptation in Children with Cerebral Palsy.” *American Society of Biomechanics*, August 13-16, 2026, Pittsburgh, PA. *Poster*

C5. **Pitts MN**, MR Ebers, CE Agresta, KM Steele. “Inferring Unmeasured Inertial Data from Sparse Sensing for Treadmill Running.” 19th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering, July 30 – August 1, 2024, Vancouver, Canada. *Poster.*

C6. **Pitts, MN**, MR Ebers, CE Agresta, KM Steele. “Leveraging Sparse Sensing to Infer Motion in Performance and Rehabilitation.” *Frontiers of Human Performance*, July 14-19, 2024, Grand Teton National Park, WY. *Podium.*

C7. **Pitts, MN**, MR Ebers, CE Agresta, KM Steele. “Reconstructing Unmeasured Inertial Data Using Sparse Sensing for Treadmill Running.” *Northwest Biomechanics Symposium*, May 17-18, 2024, Eugene, OR. *Podium.*

C8. **Pitts, MN**, CE Agresta, KM Steele. “Examining how midsole material impacts center of pressure trajectories during treadmill running.” *Northwest Biomechanics Symposium*, May 19-20, 2023, Seattle, WA. *Poster.*

TEACHING EXPERIENCE

Teaching Assistant – University of Washington

FOUNDATIONS OF MACHINE LEARNING FOR ENGINEERING — Fall 2025

- Reviewed and graded Python code and writing assignments ranging from probability and statistics to deep neural networks

SYSTEM DYNAMICS — Winter 2025

- Created review material to support problem-solving techniques and presented in office hours and discussion sections
- Led weekly lab demonstrations for mechanical, electrical, and fluid systems
- Graded homework and exams

KINEMATICS AND DYNAMICS — Winter 2023, Winter 2024

- Organized and led weekly review sessions to practice topics discussed in lecture
- Presented lectures on Newton’s laws (2023), energy (2023), and rigid body motion (2024)
- Graded homework and exams

Teaching Assistant – Catapult Engineering Academy — Aug 2018 – May 2021

INTRODUCTION TO CAD, INTRODUCTION TO CODING

- Created schedules, modified and graded assignments and tests, and answered questions for online SolidWorks and MATLAB courses for high school students

Invited Lectures – University of Washington

Musculoskeletal biomechanics, Mechanical Engineering January 2026

Structure, function, and measurement of skeletal muscle

Engineering Concepts, Prosthetics and Orthotics April 2026

Material properties and failure modes of prosthetics and orthotics for clinical evaluation

LEADERSHIP & SERVICE

Research Mentor, University of Washington Jan 2025 – Present

David Green, undergraduate – design and development of real-time oxygen consumption estimation tool

Community and Activism Chair, Black Graduate Student Association, University of Washington 2024

A Vision for Electronic Literacy and Access, University of Washington 2022-2024

President of Pi Tau Sigma, Auburn Chi Chapter, Auburn University 2021

Instated initiatives to promote engagement and community, led to national Best Chapter nomination

AWARDS & HONORS

Gatzert Child Welfare Fellowship, UW 2026

One quarter of funding for the writing of a dissertation that supports the lives of children with disabilities

NINDS Research Supplement to Promote Diversity March 2024

GEM Associate Fellow 2024

Mechanical Engineering Excellence Award, UW September 2022

O'Neal Austin Best Student Award, System Dynamics Fall 2021

Awarded each semester to the highest performing student in core mechanical engineering courses

Spirit of Auburn Presidential Merit Scholarship August 2018

100 Women Strong Scholarship August 2018